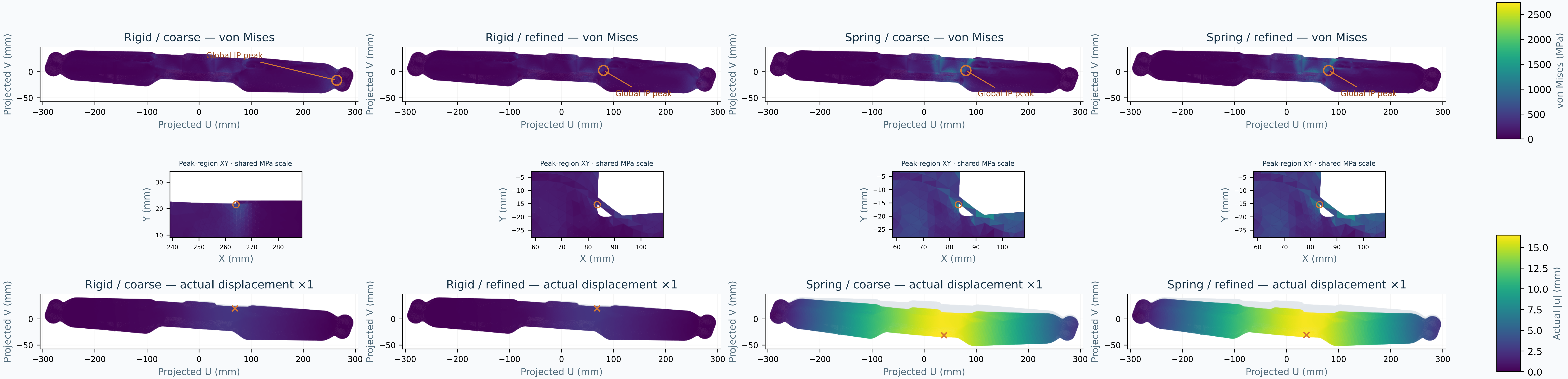
993 TURBO · SAME d64 CAD · FOUR CLOSED LINEAR CASES

Coarse versus refined: stress and displacement

Fz = −4 000 N · My = +850 000 N·mm · common camera, mm axes and field scales

Coarse: 85 767 T10 / 159 612 nodes · refined: 87 256 T10 / 161 826 nodes · true deformation ×1 · BREP d64a7faf8455…

Rigid: 4 552 nodes fixed in XYZ. K1000: 1 000 N/mm TOTAL per eye and axis, distributed over 2 278 / 2 274 nodes.



Case	Global IP peak (MPa)	Boundary max (MPa)	All-node |u| (mm)	Element / IP	Reference IP XYZ (mm)
Rigid / coarse	600.659	600.659	2.709626	35381/2	264.01521, 21.48504, -11.31000
Rigid / refined	925.773	925.773	2.710864	85784/2	83.42166, -15.36681, 13.68569
Spring / coarse	1548.152	1548.152	16.478757	81008/3	83.30876, -15.65391, 13.52606
Spring / refined	2740.902	2740.902	16.482678	85784/2	83.42166, -15.36681, 13.68569

Stress colours: maximum of 4 IPs in each boundary-adjacent element, without nodal extrapolation. Orange circles: every global IP peak.

Linear C3D10 corner facets; actual corner displacement ×1. Grey: reference shape. Crosses: every all-node maximum of |u|.

Displacement colours use the largest of each facet’s 3 corner values; the global maximum also includes midside nodes.

Same d64 BREP, two actual meshes. Assumed linear loads/material; no strength allowable or stress convergence established.

Original generated-volume refusal retained; distinct serialized-reference/H2 checks passed. Orthographic view 8° / 22°.